v0.7 Decoder: rubric v1 scored against the original deliverables

Date: 2026-10-16. Scope: nine existing attempts, seven scientific submissions; see standard-answer.md for the original prompt hash.

1. Nature and limits of the scoring

This is a post hoc local 100-point scale created at the user's request in this round, not official SciCode weights or an evaluator frozen before the original runs. Existing pass/fail/unknown judgments and original score files are retained. Quantification in this version helps compare completion; a total-score threshold does not substitute for scientific correctness or formal difficulty calibration. The A/B difficulty conditions remain incomplete.

The main score follows the original R1-R4 and explicit deliverables, totaling 100 points. The original problem permits unresolved with valid bounds, so honest preservation of unknowns does not itself lose points. The problem specifies no algorithm; correct alternatives are accepted equally. A smaller valid R3 N0 is compared separately as strength of the result; a valid but larger N0 is not treated as a failed proof.

General proportional anchors: 0=missing or a core error; approximately 25%=fragments only; approximately 50%=substantial completion with a clear gap; approximately 75%=mostly complete with a specific reservation; 100%=sufficient evidence for the requirement. Half-points or quarter-points are allowed, with reasons and evidence locations. Do not deduct twice for the same defect: for example, claims about approximation bias are primarily scored under A3, while D items assess only the required small-size API.

Infrastructure failures are unknown/NA, not scientific zeroes. An incomplete submission must also be distinguished from the absence of a reference answer. Each score records what the submission actually stated, what independent checks support, and what remains unknown. A new proof supplied by reviewers cannot be represented as work completed in the AI's original submission.

2. Main scoring table: 100 points

ID	Maximum	Requirement	Evidence for full credit and main deduction grounds
A1	5	R1 coverage	All 21 O0/O1/O2 entries for N=2,...,8, with intervals/uncertainties or justified unresolved; deduct corresponding coverage for missing entries
A2	8	R1 basis for phase classification	Increasing two-dimensional sizes, probability coverage, statistics, and extrapolation sufficiently support the classifications; slow decline/apparent plateaus alone do not determine zero/positive limits; a theorem can directly support a phase region without mandatory repeat MC
A3	8	R1 uncertainty control	Statistical, finite-size, and approximation errors reported separately, with justified numerical error estimates; chi differences are not presented as true-bias upper bounds. Do not repeat deductions here for gaps in central-region limiting-risk inference already scored under A2
A4	5	R1 figure and classification semantics	Distinguish rigorous sufficient bounds, rigorous undecodability, numerical boundaries, and unknowns; bound roots are not true transitions; figures/tables agree with answers
A5	4	R1 structural consistency	Phase classifications respect information inclusion/equivalence classes, inheriting subsets where necessary; handle the N4 anomaly and asymmetry appropriately; individual sufficient bounds need not all be strictly nested

ID	Maximum	Requirement	Evidence for full credit and main deduction grounds
B1	4	R2 model/geometry	Correct noise, conditional independence of observations, boundaries, and binary success condition; no access to hidden information or replacement of the noise channel
B2	4	R2 information order	O2-to-O1-to-O0 degradation, risk inequalities, and inclusion of decodable sets; cross-N ordering must actually be proved if used
B3	5	R2 charge equivalence	O1(N2)=O0, N3=N6, N5=N7=N8 reveal full integer q, and N4 is a separate class; only relations relied on by the answer need proof; incorrect all-N monotonicity is not rewarded
B4	2	R2 special ranks	Correct irrep probabilities and small-N exceptions where used, particularly SU2 and N4 determinant coincidences; avoid the incorrect identification O2(N2)=O0
C1	3	R3 statement	Explicit integer N0, every N>=N0 and every p, and the risk conclusion as L tends to infinity; domain and endpoints complete
C2	5	R3 witness/decoder	A valid auxiliary rule and feasible witness, or an equivalent valid direct Bayes/posterior argument; the implication from failure to a spanning component and its required conditions is sound; hiding is not mistaken for sufficiency
C3	6	R3 correlations	Valid treatment of shared-edge orientations, side edges/chords, and conditional independence of readouts; effective alternatives such as matrices, stochastic coarse-graining, and grouped Cauchy-Schwarz are accepted equally
C4	4	R3 exponential bound and certification	Fixed-path probability, path count, length lower bound, strict all-p factor<1, and finite prefactor; numerical optimization alone without a certified global maximum warrants a reservation
C5	2	R3 scope of conclusions	Sufficient N0 is clearly not the minimum/a phase boundary, and r>=1 does not mean undecodability; evidence is auditable; smaller N0 separately records result strength
D1	6	R4 joint partition sum	Preserve shared orientations at the two ends of the same edge, with correct local probabilities and logical sectors; do not independently orient vertices or replace sector posterior with MAP
D2	3	R4 exact algorithm	Exact or certified as required for L<=4,N<=8; explain contraction/enumeration and complexity; approximations at larger L do not compromise the small-size commitment
D3	5	R4 independent verification	Compare against independent enumeration/reference without reusing the same core error; evidence supports the required precision and coverage; retain the limitations of existing external heldout checks
D4	3	R4 numerical range	Positive-probability records are not misclassified as zero through underflow; normalization/log/high precision and endpoints work; distinguish platform differences from original-environment gaps
D5	3	decode interface	Return a valid edge set with the correct syndrome and a logical decision compatible with the posterior rule used; exceptions and invalid outputs can be located
E1	4	answers.json	Structure, 21 entries, endpoint uncertainties/unresolved forms, and R3_N0 agree with the report
E2	3	Report/figures	Phase diagram, R2/R3 proofs, evidence, and limitations are provided in locatable files; scientific points are not deducted for personal aesthetic preferences
E3	3	Source/reproduction/resources	API, dependency environment, commands, seeds/resource descriptions support reproduction; the 8CPU/32GiB/no-GPU requirements are documented
E4	5	Data/unconditional accounting	Raw data, seeds, stopping rules, trial denominators, and exceptions/invalid outputs counted as failures; reserved slots distinguished from actual execution

Subtotals: R1=30, R2=15, R3=20, R4=20, deliverables/reproducibility=15.

3. Scientific pass conditions are not replaced by the total score

R1: exact unknowns may remain unresolved, but high scores elsewhere cannot turn a key definite phase classification without sufficient evidence, or an approximation difference presented as a true error bound, into a pass.

R3: a fatal gap in the core witness, correlations, or strict all-p bound means this requirement is unmet; a valid larger N0 still passes. Arithmetic evidence in the original submission is listed separately from independent repairs after review.

R4: the small-size precision and interface required by the problem must be met. Passing ordinary heldout checks does not establish robustness over the entire positive-probability input domain; stress-test gaps should identify engineering and platform aspects. Such readily repairable problems should not alone be treated as failures of the scientific judgment sought by SciCode.

Whole task: review each main requirement, allowing qualified/partial/unknown outcomes; no post hoc automatic pass threshold of 80 points. Without guided B, there is no formal A/B difficulty conclusion.

4. Discussion-detail diagnostics: no extra points within the 100-point score

The following address whether the AI handled each detail properly. Each run is marked E=explicitly correct discussion, I=implicit or only partly developed, M=no discussion found, X=explicit error/inference with insufficient evidence, NA=not required by a valid alternative method, U=no scientific submission. M does not automatically mean failure of the original task; X requires a specific claim and location.

ID	Diagnostic question	Relation to the original task
Q01	D_r(N) is the full-p set; more information cannot shrink it but need not strictly enlarge it	Core R2 meaning
Q02	N5 already recovers full integer q; exact saturation for N>=7, with N6 an exception	R2 core
Q03	mod4 and mod6 do not contain one another; no all-N ordering	Required when making cross-N comparisons
Q04	Non-nested standalone sufficient bounds do not imply that true phases violate information ordering	R1 interpretation
Q05	Transfer reliable Z4 decodable subsets to full integer q	R1/R2 consistency
Q06	Distinguish Z4 confusion of pure triples with singles from mixed triples with singles	Feature mechanism
Q07	High-p active-component orientation identity and the scope of its proof	Optional enhanced analysis
Q08	O1 cannot see neutral pairs, while O2 hiding becomes rarer with N	R3/phase-diagram mechanism
Q09	Separate binary X,C from latent sigma orientation; sigma is not given to the decoder	Core model
Q10	Difference-component self-avoidance follows from degree 0/2, not physical repulsion	Required when using a path proof
Q11	M is the minimum compatible count; only 11 forces true hiding, while 01/10 leaves the count unchanged	Required when using this coarse constraint
Q12	Hiding is necessary, not sufficient, and applies to the specified auxiliary decoder	Required when using a hiding proof
Q13	Decoder score/minimum or maximum edge count/Chernoff conditions enter the probability bound	Required for the corresponding proof approach

ID	Diagnostic question	Relation to the original task
Q14	Correct explanation of orientation-matrix states and every element	Applies to the direct matrix approach; NA for alternatives
Q15	Occupation and orientation matrices have different state spaces	Applies to the two-matrix approach; NA for alternatives
Q16	Shared off-path edges/chords cannot be independently resampled vertex by vertex	Core correlations; valid avoidance can receive E
Q17	R*<=R_aux compares average risks, and the exponential bound is not a critical exponent	Core proof conclusion
Q18	The partition function preserves global orientations; small-size exact computation is separate from large-size MPS	R4 core
Q19	Separate phase estimates/sufficient bounds/unknowns and statistical/finite-size/approximation uncertainties	R1 core
Q20	Every exception counts as a failure; reserved slots are not completed trials	Explicit deliverable requirement
Q21	Illustrate sigma, F, and endpoint summation element by element	New teaching request in this discussion; not required by the original task

Using a,b for both matrix indices and arrival counts in this chat was a notation error in the review explanation; it must not be recorded as a shared error in the seven AI submissions. The reference answer uses alpha,beta and n_V,n_barV instead.

5. Preservation and review

scores-v1.json retains weights, item scores, reasons, evidence, and diagnostics; scores-v1.csv facilitates sorting; scorecards.md supports human review. All original run directories are read-only. New review conclusions and differences from existing reviews are listed separately, without overwriting the old scores.json. These detailed weights can be frozen in advance for a future batch of new tests; for this batch they remain a post hoc detailed review.

Aspect scores and discussion-detail audit for nine attempts, v1

This is a post hoc local completion rubric, not official SciCode weights or a prefrozen evaluator. Totals do not replace the R1-R4 scientific pass conditions. Seven scientific submissions and two API failures are retained.

1. Overview

Run	R1 /30	R2 /15	R3 /20	R4 /20	Delivery /15	Total /100	N0	Scientific assessment
Fable03	26	15	19.5	18	14	92.5	19	Partial completion: gaps in central-phase evidence and numerical range
Opus01	22	15	20	18	14.5	89.5	15	Partial completion: gaps in central extrapolation, approximation guarantees, and numerical range
Opus02	22	15	20	18	14.5	89.5	15	Partial completion: gaps in central extrapolation, approximation guarantees, and numerical range
Opus03	22	15	19	18	14.5	88.5	15	Partial completion: as above, with additional caveats on the original all-p arithmetic certification
GPT01	30	15	20	20	15	100	13	Provisional pass with the conservative answer permitted by the original task
GPT02	30	15	20	20	15	100	13	Provisional pass with the conservative answer permitted by the original task
GPT03	29	15	20	20	15	99	14	Provisional pass with caveats: numerical all-p extrapolation at small N
Fable01	NA	NA	NA	NA	NA	NA	NA	API output limit; not scientifically assessable
Fable02	NA	NA	NA	NA	NA	NA	NA	API output limit; not scientifically assessable

The 100 for GPT01/02 means that the limited conclusions permitted by the original task satisfy this rubric, not that the complete physical phase diagram is solved. All seven produce high-quality core theory and ordinary posteriors, so partial submissions can still receive high completion scores; totals do not obscure key scientific gaps. Smaller valid N0 values are reported separately as stronger results.

R3 is scored as submitted: 0.5 and 1 point are withheld for Fable03 and Opus03, respectively, for incomplete all-p arithmetic certification. Existing independent rational verification supports their theorems; their N0 conclusions have not been refuted.

2. Matrix of 23 scoring criteria

Criterion	Maximum	Fable03	Opus01	Opus02	Opus03	GPT01	GPT02	GPT03
A1 R1 coverage	5.0	5	5	5	5	5	5	5
A2 R1 basis for phase classification	8.0	4	5	5	5	8	8	7
A3 R1 uncertainty control	8.0	8	3	3	3	8	8	8
A4 R1 figure and classification semantics	5.0	5	5	5	5	5	5	5
A5 R1 structural consistency	4.0	4	4	4	4	4	4	4
B1 R2 model/geometry	4.0	4	4	4	4	4	4	4
B2 R2 information order	4.0	4	4	4	4	4	4	4
B3 R2 charge equivalence	5.0	5	5	5	5	5	5	5
B4 R2 special ranks	2.0	2	2	2	2	2	2	2
C1 R3 statement	3.0	3	3	3	3	3	3	3
C2 R3 witness/decoder	5.0	5	5	5	5	5	5	5
C3 R3 correlations	6.0	6	6	6	6	6	6	6
C4 R3 exponential bound and certification	4.0	3.5	4	4	3	4	4	4
C5 R3 scope of conclusions	2.0	2	2	2	2	2	2	2

Criterion	Maximum	Fable03	Opus01	Opus02	Opus03	GPT01	GPT02	GPT03
D1 R4 joint partition sum	6.0	6	6	6	6	6	6	6
D2 R4 exact algorithm	3.0	3	3	3	3	3	3	3
D3 R4 independent verification	5.0	5	5	5	5	5	5	5
D4 R4 numerical range	3.0	1	1	1	1	3	3	3
D5 decode interface	3.0	3	3	3	3	3	3	3
E1 answers.json	4.0	4	4	4	4	4	4	4
E2 Report/figures	3.0	3	3	3	3	3	3	3
E3 Source/reproduction/resources	3.0	3	3	3	3	3	3	3
E4 Data/unconditional accounting	5.0	4	4.5	4.5	4.5	5	5	5

3. The 21 details discussed

E=explicit and correct; I=implicit/partially developed; M=not found; X=specific error or insufficiently supported inference; NA=unneeded for a valid alternative route; U=no scientific submission. Teaching diagrams were requested later and do not incur retrospective deductions.

Detail	Fable03	Opus01	Opus02	Opus03	GPT01	GPT02	GPT03
D_r(N) is the full-p set; more information cannot shrink it but need not strictly enlarge it	E	E	E	E	E	E	E
N5 already recovers full integer q; exact saturation for N>=7, with N6 an exception	E	E	E	E	E	E	E
mod4 and mod6 do not contain one another; no all-N ordering	E	E	E	E	E	E	E
Non-nested standalone sufficient bounds do not imply that true phases violate information ordering	I	I	I	I	E	E	I
Transfer reliable Z4 decodable subsets to full integer q	I	I	I	I	E	E	I
Distinguish Z4 confusion of pure triples with singles from mixed triples with singles	I	I	I	E	E	I	I
High-p active-component orientation identity and the scope of its proof	M	M	M	E	E	M	M
O1 cannot see neutral pairs, while O2 hiding becomes rarer with N	E	E	E	E	E	E	E
Separate binary X,C from latent sigma orientation; sigma is not given to the decoder	E	E	E	E	E	E	E
Difference-component self-avoidance follows from degree 0/2, not physical repulsion	E	E	E	E	E	E	E
M is the minimum compatible count; only 11 forces true hiding, while 01/10 leaves the count unchanged	I	I	I	I	E	E	E
Hiding is necessary, not sufficient, and applies to the specified auxiliary decoder	E	E	E	E	E	E	E
Decoder score/minimum or maximum edge count/Chernoff conditions enter the probability bound	E	E	E	E	E	E	E
Correct explanation of orientation-matrix states and every element	NA	NA	NA	NA	E	E	NA
Occupation and orientation matrices have different state spaces	NA	NA	NA	NA	I	E	NA
Shared off-path edges/chords cannot be independently resampled vertex by vertex	E	E	E	E	E	E	E
R*<=R_aux compares average risks, and the exponential bound is not a critical exponent	E	E	E	E	E	E	E
The partition function preserves global orientations; small-size exact computation is separate from large-size MPS	E	E	I	E	E	E	E

Detail	Fable03	Opus01	Opus02	Opus03	GPT01	GPT02	GPT03
Separate phase estimates/sufficient bounds/unknowns and statistical/finite-size/approximation uncertainties	I	X	X	X	E	E	E
Every exception counts as a failure; reserved slots are not completed trials	X	X	X	X	E	E	E
Illustrate sigma, F, and endpoint summation element by element	M	M	M	M	M	M	M

NA does not mean correlations were ignored: Fable/Opus/GPT03 use valid alternative proofs. I/M also does not automatically mean task failure; the high-p Z4 identity and cell-by-cell teaching diagrams are enhanced explanations. For Q20, read the run-specific evidence to distinguish latent implementation defects from exceptions that actually occurred.

4. Corrections to earlier assessments

All seven correctly analyze complete integer q at N5 and saturation for N>=7; this is not a widespread omission. GPT01 and Opus03 give the Z4 active-component identity. Nonnested individual sufficient bounds do not imply that the true optimal phase regions violate information inclusion.

The earlier REPORT characterized the three Opus central-region explanations too broadly: the Gaussian/BKT risk-lower-bound exclusion argument explicitly belongs to Opus02/03. Opus01 should be described as a physical explanation of typical charges pinning the clean-loop critical phase, together with slow-drift extrapolation. This corrects the attribution without revising the historical assessment.

All seven pass all 210 ordinary posterior/correction cases each: 140 original R4 cases and 70 O0 controls. GPT passes all 18 extremely-low-p stress inputs; the other four fail locally. The Linux/macOS longdouble range difference for Opus02 remains a caveat.

The Fable03 exception fallback to posterior=.5 would undercount failures, but its archived exception count is zero, making this a latent defect. Opus records hard failure as 1 but still uses .5 for posterior risk. GPT03 has 340600 archived slots, including 200 reserved conservative failure slots; these cannot all be called completed physical trials.

Appendix: evidence for deductions and caveats

All 161 scientific-submission scores, 21 discussion diagnostics, and full absolute paths are retained in scorecards.md, scores-v1.json, and score-details-v1.csv. Items affecting scores or conclusions are collected below.

Fable03

- A2: 4/8.0. Provides exact-posterior data at increasing two-dimensional sizes, but the positive-risk central charge phase is assigned from slow drift/positive-limit fits at L<=12; a very long crossover is not excluded. Source: `anthropic__claude-fable-5-1/run_03/report.md:213`.
- C4: 3.5/4. The original all-p rigorous arithmetic certification has caveats; existing independent rational verification supports the theorem. Source: `anthropic__claude-fable-5-1/run_03/docs/theory.md:224`.
- D4: 1/3. Passes 210/210 routine cases; fails 18 extremely low-p positive-probability stress cases. Ordinary stepwise normalization is present, but local products or edge weights can still underflow. Source: `anthropic__claude-fable-5-1/run_03/src/model.py:198`.

- E4: 4/5. Preserves seeds, fixed job counts, resources, raw arrays and denominators; exceptions become P0=.5 and aggregation gives both risk and hard tie .5, whereas the task requires failure. data/summary.csv has total nfail_exc=0, so this is a latent defect with no observed effect on archived estimates. Source: `anthropic__claude-fable-5-1/run_03/src/run_mc.py:56`.
- The exception-accounting defect is latent: the archived summaries total zero exceptions, and no change to the archived phase-diagram values was found.

Opus01

- A2: 5/8.0. Two-dimensional sizes reach about L48 and the diagnostics are useful; the zero central charge limit and charge-pinning explanation for the clean-loop phase still do not exclude alternative positive limits. Source: `anthropic__claude-opus-5-5/run_01/report.md:326`.
- A3: 3/8.0. Substantial chi comparisons and error classification are provided, but agreement between two approximations is not an upper bound on bias relative to the true posterior. Source: `anthropic__claude-opus-5-5/run_01/report.md:82`.
- D4: 1/3. Passes 210/210 routine cases; fails 18 extremely low-p positive-probability stress cases. Ordinary stepwise normalization is present, but local products or edge weights can still underflow. Source: `anthropic__claude-opus-5-5/run_01/solution.py:71`.
- E4: 4.5/5. Raw data, seeds, fixed-block stopping rules, resources and denominators for all attempts are recorded; bad hard failure is 1, but the reported posterior-min risk is still .5 rather than 1 on exception. Source: `anthropic__claude-opus-5-5/run_01/src/mc.py:146`.

Opus02

- A2: 5/8.0. Two-dimensional sizes reach about L96; the Gaussian/BKT heuristic is not proved sufficient to exclude every nondecodable phase of the specified posterior model. Source: `anthropic__claude-opus-5-5/run_02/REPORT.md:184`.
- A3: 3/8.0. Promotes the difference between chi and larger chi to an induced-bias upper bound without controlling the error of the latter. Source: `anthropic__claude-opus-5-5/run_02/REPORT.md:301`.
- D4: 1/3. Passes 210/210 routine cases; fails 18 extremely low-p positive-probability stress cases. Ordinary stepwise normalization is present, but local products or edge weights can still underflow. macOS longdouble has a different range from the original Linux environment, so not all macOS failures can be attributed directly to the original environment; however, p/2 in pri rounds to zero in Python float before conversion to dtype, a cross-platform issue at the smallest positive float. Source: `anthropic__claude-opus-5-5/run_02/hexsym/api.py:45`.
- E4: 4.5/5. Raw data, seeds, fixed-block stopping rules, resources and denominators for all attempts are recorded; bad hard failure is 1, but the reported posterior-min risk is still .5 rather than 1 on exception. Source: `anthropic__claude-opus-5-5/run_02/hexsym/sim.py:49`.
- The extremely-low-p numerical gap includes the longdouble range difference between the original Linux environment and the macOS review; loss of the smallest subnormal when computing p/2 first in Python float is independent of that difference.

Opus03

- A2: 5/8.0. Two-dimensional sizes reach about L64; slow decline and Gaussian/BKT exclusion do not establish a zero central charge limit. Source: `anthropic__claude-opus-5-5/run_03/report.md:239`.
- A3: 3/8.0. Larger chi can share the same bias; empirical convergence differences do not provide the claimed guarantee on true bias. Source: `anthropic__claude-opus-5-5/run_03/report.md:355`.
- C4: 3/4. The original all-p rigorous arithmetic certification has caveats; existing independent rational verification supports the theorem. Source: `anthropic__claude-opus-5-5/run_03/docs/proofs.md:199`.

- D4: 1/3. Passes 210/210 routine cases; fails 18 extremely low-p positive-probability stress cases. Ordinary stepwise normalization is present, but local products or edge weights can still underflow. Source: `anthropic__claude-opus-5-5/run_03/symdec/api.py:43`.
- E4: 4.5/5. Raw data, seeds, fixed-block stopping rules, resources and denominators for all attempts are recorded; bad hard failure is 1, but the reported posterior-min risk is still .5 rather than 1 on exception. Source: `anthropic__claude-opus-5-5/run_03/symdec/mc.py:121`.

GPT01

All main criteria of this rubric are satisfied. The central phase region remains unknown, as in the reference answer; the full phase diagram is not thereby solved.

GPT02

All main criteria of this rubric are satisfied. The central phase region remains unknown, as in the reference answer; the full phase diagram is not thereby solved.

GPT03

- A2: 7/8.0. Exact posteriors at increasing two-dimensional sizes support a numerical all-p assignment for small-N O2; finite grids and late crossovers remain caveats, and this is not an all-p theorem. Source: `openai__gpt-6-astra/run_03/report.md:3`.
- The 340600 archived slots include 200 reserved failure slots whose execution is unverified; the report interprets them with this qualification.

Appendix: version and review scope

Original prompt SHA-256: `d89b765a4af1e50b2824d40d0c2d8590e722202219411bd49f5ea7d63c64d6a9`. The task is unchanged, with no new model runs or production phase-diagram experiments. Existing independent oracles, 210 ordinary checks, 18 stress checks, and large-N rational certificates are reused; this review adds the rubric, detailed deliverable audit, and certification of reference safe endpoints.

The scores are not a formal SciCode A/B difficulty conclusion. The full official template remains unverified; local weights are not represented as official rules. Totals measure completion, while key scientific pass conditions remain individually stated.